Camera Ready Summary

Conference Name

12th IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering 2026

Paper ID

518

Paper Title

OrdinalFed: Federated Learning with Ordinal-Aware Aggregation for Diabetic Retinopathy Grading

Abstract

Diabetic Retinopathy (DR) is a leading cause of preventable blindness. With federated learning (FL), hospitals can train one DR grading model together without sharing retinal images. When hospitals see very different patients, however, FedAvg can collapse. In this paper, OrdinalFed is studied. It has two parts. The first is an OrdinalLoss that penalises a wrong grade by its squared distance from the true grade. The second is a QWK-weighted aggregation: the server scores each client model by its Quadratic Weighted Kappa (QWK) on 50 held-out images, and over the rounds it moves the aggregation weights from image counts towards these scores. All experiments on the DDR dataset were repeated with three seeds and scored on the held-out test split. Under extreme label skew (Dirichlet parameter 0.1), FedAvg collapsed in every seed, with a worst-round test QWK of 0.378. OrdinalFed raised the worst round to 0.514, with no change at the hospitals and about 1.2 s of extra server time per round. The ablations show where this gain comes from: small hospitals get more weight. Equal client weights did even better (0.630), while the QWK scores and the OrdinalLoss added no measurable benefit. FedProx was the most stable method (0.641), and OrdinalFed with FedProx's proximal term came close to it (0.627). On the IID and realistic multi-site partitions, all methods were within 0.01 QWK of each other. On the external APTOS 2019 dataset, all models scored 0.597–0.658 QWK, close to the centralised model (0.641).

Keywords—Diabetic Retinopathy, Federated Learning, Non

IID, Ordinal Loss, QWK-Weighted Aggregation, EfficientNet-B3, Medical Imaging

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